

Big Data Analytics and Machine Learning for Smart Agriculture

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Modern technologies are continuously entering every aspect of our lives. Today, we can no longer do without smartphones, intelligent applications, and devices that significantly contribute to the improvement and comfort of our lives. In addition to functions related to the acceleration of everyday activities and household processes, aspects related to artificial intelligence and machine learning support industry and the production of goods and services. One such branch of the economy is agriculture, the primary goal of which is to meet human nutritional needs. Taking actions against hunger and striving for greater yields and better-quality harvests are some of the most important goals of current agriculture. The era of industrialization and the implementation of mechanization, including the mechanization of agriculture, is followed by the digital era [1,2].

It is widely believed that digitization is revolutionizing the world. However, it should be noted that it will not replace machines and devices in agriculture, but it can modify and optimize plant and animal production process. Since the first edition of “Big Data Analytics and Machine Learning for Smart Agriculture” in 2023, many technologies have been implemented into agricultural technology, changing it from Agriculture 4.0 to 5.0 [3,4].

Over the past few years, big data analysis and machine learning have revolutionized the management of agricultural systems and farms [5–7]. The large amounts of data collected daily during the observation of vegetation processes, the harvesting of crops, and their processing into food have contributed to the new knowledge published in this Special Issue.

The published works reveal how collecting and storing data contributed to the creation of an algorithm for detecting weeds, which was created based on YOLOv8. He’s team presented an improvement in the basic network by adding the so-called attention mechanisms and using dynamic convolution [8]. In another paper, the authors used deep learning, also based on YOLO, RT-DETR, and Mask-RCNN technologies, to detect and classify the ripeness of blueberries. Aguilera’s team emphasized the importance of model optimization, and their results suggest that new algorithms and their correlation with empirical studies increase the effectiveness of the created systems, thus increasing the effectiveness of crop monitoring [9].

Artificial intelligence methods based on analyzed data can also be used to create a fertilization recommendation system, which optimizes the use of agricultural production means. Musanase et al. revealed that implementing such solutions in precision agriculture can not only increase yields, but also reduce fertilizer losses and contribute to the implementation of sustainable agricultural practices [10]. Properly collected and processed data allow for the development of empirical system models supported by AI algorithms for forecasting industrial hemp seed yields. After considering data on climatic conditions,



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agrotechnics, and seed quality, Sieracka and her colleagues created predictive models to assess crop efficiency, determining the factors that have the greatest impact on seed efficiency and yield [11].

Xie's team converted empirical data into digital data, which can be used to develop algorithms that identify pathological changes in fruits and vegetables, such as peppers. Systems based on AI solutions can improve the precision and efficiency of diagnosing plant diseases, which is important in Agriculture 5.0 [12]. Similarly, Bai's and Amin's teams described the use of AI techniques and technological achievements such as drones and other devices that not only identify plant diseases, but also detect pests, both of which cause plant damage and reduce the quantity and quality of crops [13,14].

Nazir et al. indicated that models supporting the identification of diseases occurring on potato leaves and allowing for the classification of disease stages can generally affect the response time and the application of appropriate measures, thus reducing crop losses [15]. Leaf diseases are often the first prognosticator of a more complex problem related to the proper vegetation of plants. Tomatoes are a popular plant characterized by beneficial nutritional and health-promoting properties for humans. By collecting appropriate data and processing them into digital form, Ullah's team showed that it is possible to develop a classification model defining pathological disease changes in these plants [16].

Guava diseases (leaf blight) can also be detected based on the identification and classification of leaves. Depending on the plant and the type of problem and its complexity, various technologies can be used, such as convolutional networks or deep learning methods, e.g., those developed by Mumtaz et al. [17]. The basis for plant vegetation and crop quality is the environment in which the plant grows, especially the quality of the soil. In this respect, Shahare's team showed that data can also be used to develop appropriate models based on machine learning methods to assess and forecast the activity of soil enzymes, which are key to biological processes occurring in the soil, to help farmers optimize agricultural production [18].

Plant production, including field or greenhouse crops, is just one branch of modern agricultural production. The second largest aspect of agricultural production is the animal production branch. Artificial intelligence can be used to optimize meat production, e.g., beef from dairy cattle, as shown by Addis et al. Such systems can increase production efficiency regarding the use of animals for dairy or meat production and in providing unified food to consumers [19].

Forecasting and advisory systems are also based on artificial intelligence algorithms, which are based on large data sets, allowing for broad diagnosis and the prediction of failures in increasingly popular smart farms. Choe's team showed that such systems can be used to detect irregularities in data obtained from sensors and can predict potential failures of agricultural equipment [20].

Notably, machine learning models, neural models, or ordinary linear forecasting models cannot be created without previously collected data that have been appropriately processed and adapted for analysis. It is very difficult to compare research results concerning the same object. Data are collected using different devices and sensors. Additionally, there are varying frequencies of data collection, and devices have different operating conditions. Similarly, the varieties of measured plants and the fields or buildings in which livestock are kept are different.

This is a significant problem in establishing a methodology and indicating objective conclusions. We are surrounded by a multitude of data. We continuously collect and try to systematize data. It is challenging to not only collect and store data for a long time, but also to effectively process and analyze them to obtain valuable information. Modern IT tools allow us to systematize data, discover patterns, and generate new scientific

knowledge, supporting innovations in digital agriculture. However, data are the basis, and they represent the most time-consuming and cost-intensive part of the research process. Nevertheless, obtaining a large amount of data will allow for the creation of new methods and technologies supporting agricultural production.

In this Special Issue, particular emphasis is placed on big data in agriculture and machine learning, i.e., on the collection, management, and analysis of large data sets, analysis and prediction, decision support systems, and automation based on AI IoT methods. It also focuses on the integration of sensor networks and intelligent monitoring systems, which allows for the transition to a new era, Digital Agriculture 5.0, in which automation, robotics, and artificial intelligence support modern precision agriculture and enable sustainable development through resource optimization, loss reduction, and improved efficiency.

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